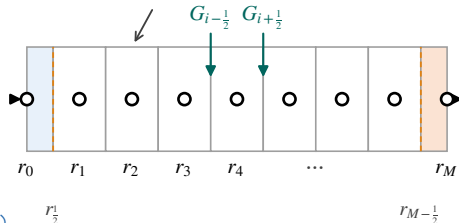


(b) 一维径向求解域与节点式有限体积 (图示 $M=8$)

$$V_i = \pi(r_{i+\frac{1}{2}}^2 - r_{i-\frac{1}{2}}^2) \quad (\text{单位轴长})$$



中心: $\partial T / \partial r = \partial C / \partial r = 0$ (对称)

表面: $-k \partial_r T = h(T - T_\infty),$

$-D \partial_r C = h_m(C - C_\infty)$

(a) 物理模型: 圆柱药材与第三类边界条件

